MILITARY SPECIFICATION

HEAD ASSEMBLY, EMPTY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of head assembly entitled Head Assembly, Empty which is used on the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-5021	- Tests: Aircraft Welding Operators Certification.
MIL-C-6021	- Castings, Classification and Inspection of (For Aeronautical Applications).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

ORDNANCE CORPS

8148433	- Head Assembly, Empty.
8821965	- Fuze, Training.
8824934	- Case Shipping, XM136E1 for Adaption Kits and Warhead Section, Packing and Marking.

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PROCESS SPECIFICATIONS

PICATINNY ARSENAL

PPS-R-116 - Riveting, General Specification For.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The head assembly shall comply with all requirements specified on the drawing (dwg) 8148433, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Castings. All castings shall be sound, dense, and free from casting defects such as irregular shrinkage porosity or sponginess, large spherical blow holes, intercrystalline cavities or microshrinkage, surface depression or sinks, and cracks. All castings shall be cleaned by sandblasting, wire brushings, or other equally effective means when tested as specified in 4.4.1.

3.5 Weight. The weight of the empty head assembly shall be approximately 163 + 3.3 pounds.

3.6 Paint adhesion. Painted items, specimens, or test panels shall show satisfactory paint adhesion. The specimen shall not exhibit bare metal or underlying coating in any spot exceeding 1/8 inch average diameter or more than five specks 1/16 inch or smaller average diameter when tested as specified in 4.4.2.

3.7 Welding. Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion, penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of good practice on the part of the operator. All welding fluxes, scale, weld spatter, guide, and basic solutions shall be removed prior to application of any finish coat.

3.7.1 Welding operators. All fusion welding shall be performed by operators who are currently certified as Class "B" welders in accordance with MIL-T-5021.

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3.8 Riveting. Riveting shall be as specified in Process Specifiecation RPS-R-116.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 2 complete assemblies (see dwg 8148433) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Castings (see 3.4)		X
3. Paint adhesion (see 3.6)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material, one manufacturer of components, one production method, and one manufacturing operation. The entire lot shall be submitted as a unit for inspection.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Bolt, nose tip (see dwg 8148394 covering a detail of dwg 8148225).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pitch diameter, max	Gage
	102. Major diameter, max	Gage
	103. Total length, min	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.2 Skin assembly, nose tip (see dwg 8148225 covering a detail of dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Nose bolt hole improperly located	Gage
	102. Crack in weld	Visual
	103. Improper welding (see 3.7)	Visual
Minor:		
	201. Marking missing, incorrect or unidentifiable	Visual
	202. Evidence of poor workmanship (see 3.9)	Visual

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4.3.2.1.3 Fuze training, (see dwg 8821965 covering a detail of dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any part mislocated loose or damaged	Visual/Manual
102.	Inability to mate with next assembly	Visual/Manual
Minor:		
201.	Marking missing, incorrect or unidentifiable	Visual
202.	Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.4 Shell assembly, middle section (see dwg 8148434 covering a detail of dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Rivet improperly spaced	Gage
102.	Door opening improperly located	Gage
103.	Total length	Gage
104.	Rivet cracked or loose (see 3.8)	Visual/Manual
Minor:		
201.	Marking missing, incorrect or unidentifiable	Visual
202.	Evidence or poor workmanship (see 3.9)	Visual

4.3.2.1.5 Assembly, nose base (see dwg 8800662 covering a detail of dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Rivets cracked or loose (ye 3.8)	Visual/Manual
102.	Hardware loose or missing	Visual/Manual
Minor:		
201.	Marking missing, incorrect or unidentifiable	Visual

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202. Evidence of poor workmanship (see 3.9).....Visual

4.3.2.1.6 Access door assembly, pressure diaphragm (see dwg 8148258 covering a detail of dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Patch plate improperly located	Visual
	102. Patch plate loose or missing	Visual/Manual
	103. Access door poorly fitted	Visual/Manual
	104. Hardware loose or missing	Visual/Manual
Minor:		
	201. Marking missing, incorrect or unidentifiable	Visual
	202. Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.7 Head assembly, empty (see dwg 8148433).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Weight	Scale
	102. Improper location of parts	Visual
	103. Improper fit of component parts	Visual
	104. Seal missing or damaged	Visual
	105. Hardware loose or missing	Visual/Manual
	106. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Marking missing, incorrect or unidentifiable	Visual
	202. Evidence or poor workmanship (see 3.9)	Visual

4.3.2.1.8 Barrier bags, packed (see dwg 8824934).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Bag type incorrect.....Visual
102. Bag of improper material.....Visual
103. Marking missing, incomplete or unidentifiable.....Visual

Minor: None defined.

4.3.2.1.9 Box, prior to closing (see dwg 8824934).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Box improperly packed	Visual
	102. Car seals missing or improperly positioned	Visual
	103. Cushioning material of incorrect type, inadequate or missing	Visual
Minor:	None defined.	

4.3.2.1.10 Box, closed (see dwg 8824934).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Car seal missing or incorrectly positioned	Visual
	102. Board loose, broken or split	Visual
	103. Marking missing, incorrect or unidentifiable	Visual
Minor:		
	201. Incorrect weight	Scale
	202. Enamel coat of incorrect type or missing	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Paint adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Paint adhesion. Three sample head assemblies (or test panels) shall be selected from each eight hours of production and subjected to this test (see 4.4.2). If any assembly (or test panel) fails to comply with 3.6, the lot shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions designed or specified by the contractor. The contractor shall obtain approval of such designs, operating instructions and calibration instructions prior to use on the contract from Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I as specified on DI-R-1714 (see 6.1.2).

4.3.4.2 Minor characteristics and tests. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Castings. Inspection of castings shall be in accordance with MIL-C-6021.

4.4.2 Paint adhesion. The test shall be performed on the painted head assembly (or test panels) after a maximum of 24 hours drying. Water-resistant, pressure-sensitive, adhesive tape (3/4 inch width) shall be used. Press a 2-inch length of a somewhat longer piece firmly onto a flat or cylindrical surface of the specimen rubbing out all air bubbles under the tape. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape and at a rapid speed strip it from the head assembly or test panel by pulling the tape back upon itself at 180° (in such a manner that the tape is folded back to back during the procedure). Observe for bare spots where the paint is removed. Disregard flecks of paint on the tape if removal of paint has not been completely through to the metal or underlying coating.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The head assembly shall be packaged in accordance with dwg 8824934 using the specified moisture barrier.

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5.1.2 Level C. The head assembly shall be packaged in accordance with dwg 8824934 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The head assembly shall be packed in accordance with dwg 8824934.

5.3 Marking. Marking shall be as specified in accordance with MIL-STD-1169 (MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1818 dated 12 August 1960.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A024

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